

Supplementary Material

Deep Representation Learning for High-Dimensional Omics Data

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Supplementary Note S1: compute infrastructure

ESM-2 embeddings (HPC). The embeddings came off CSUC’s Pirineus3 cluster, scheduled through Slurm onto its NVIDIA H100 cards. Setting up there was not trivial. Its shared environment is read-only and PyTorch is not installed, so I had to build a conda environment of my own — torch 2.6 on CUDA 12.4, transformers 5.9 — pulling the packages through the site proxy. I also prefetched the model weights into a local cache, so the compute nodes never touch the network. A resumable, memory-mapped, half-precision script mean-pools the last hidden state for every family. The 35M model embedded all 1,447,952 families in 1 hour 18 minutes (about 317 sequences/s) and the 650M model in 4 hours 32 minutes (about 89 sequences/s), giving a 7.4 GB embedding matrix. Rather than move that matrix around, I train the sequence and fusion models on the cluster GPU (precomputing the abundance scores locally and shipping them as a small file) and only copy the small result tables back.

Local environment. Everything else — the sample-level classifiers, the abundance models — I ran on a laptop, under conda with Python 3.12, PyTorch on the Apple-Silicon MPS backend, scikit-learn, pandas and pyarrow.

Supplementary Tables

Table S1 — Trainable model architectures and parameter counts.

Experiment	Model	Architecture (layer widths)	Trainable parameters
Phenotype	Autoencoder (pathways) + LR head	466-256-64- 32 -64-256-466	276,594 (+33)
Phenotype	Autoencoder (species) + LR head	572-256-64- 32 -64-256-572	330,972 (+33)
Prioritization	Linear (abundance)	1595-1	1,596
Prioritization	MLP (abundance)	1595-256-64-1	425,089
Sequence/fusion	SeqMLP (k-mer)	400-256-64-1	119,169
Sequence/fusion	SeqMLP (ESM-2 35M)	480-256-64-1	139,649
Sequence/fusion	SeqMLP (ESM-2 650M)	1280-256-64-1	344,449
Sequence/fusion	Fusion (logistic)	2-1	3

Table S2 — All rankers on the protein-family prioritization test set (217,193 families, 1.87 % base rate). Relates to Section 4.2.

Method	AUPRC	AUROC	P@1000	enrichment@1000
Random	0.018	0.497	0.012	0.6×
Ecology score	0.075	0.746	0.210	11.2×
MetaWIBELE supervised (published)	0.064	0.703	0.198	10.6×
MetaWIBELE unsupervised (published)	0.076	0.730	0.212	11.4×
Linear (ours)	0.104	0.797	0.264	14.1×
MLP (ours)	0.142	0.832	0.339	18.2×

Table S3 — AUPRC by characterization category, abundance models. Relates to Section 4.3.

Bucket	n	positives	base rate	MetaWIBELE unsup	MetaWIBELE sup	MLP (ours)
characterized	125,750	3,302	0.026	0.096	0.079	0.167
weak	67,135	703	0.011	0.024	0.023	0.059
homology novel (no homology)	24,308	50	0.002	0.003	0.003	0.005

Table S4 — Sequence and fusion models (test set: 217,193 families, base rate 1.87 %). Relates to Section 5.2.

Method	AUPRC	AUROC	P@1000	enrichment@1000
MetaWIBELE supervised	0.064	0.703	0.198	10.6×
MetaWIBELE unsupervised	0.076	0.730	0.212	11.4×
Abundance (MLP)	0.142	0.832	0.339	18.2×
Sequence-only (k-mer)	0.182	0.875	0.351	18.8×

Method	AUPRC	AUROC	P@1000	enrichment@1000
Sequence-only (ESM-2 35M)	0.241	0.902	0.452	24.2×
Sequence-only (ESM-2 650M)	0.271	0.910	0.481	25.8×
Fusion (abundance + k-mer)	0.307	0.911	0.583	31.2×
Fusion (abundance + ESM-2 650M)	0.368	0.927	0.631	33.8×

Table S5 — Multi-seed test AUPRC (mean $\pm$ 95 % CI over 5 splits; ESM-2 35M pipeline). Relates to Section 5.5.

Method	mean AUPRC	95% CI
MetaWIBELE supervised	0.067	[0.065, 0.069]
MetaWIBELE unsupervised	0.078	[0.075, 0.081]
Abundance MLP	0.140	[0.137, 0.144]
Sequence-only (ESM-2 35M)	0.227	[0.223, 0.231]
Fusion (abundance + ESM-2 35M)	0.348	[0.341, 0.356]

Table S6 — AUPRC by characterization category (ESM-2 650M test predictions). Relates to Section 5.6.

Bucket	n	pos	base rate	MetaWIBELE unsup	Abundance MLP	Sequence ESM-2 650M	Fusion
characterized	125,750	3,302	0.026	0.096	0.167	0.301	0.403
weak	67,135	703	0.011	0.024	0.059	0.184	0.212
homology novel (no homology)	24,308	50	0.002	0.003	0.005	0.074	0.091

Table S7 — Pfam domains most enriched in the top-1000 ranked families. Relates to Section 5.8.

Pfam	Domain	top / bg count	odds ratio	FDR
PF07715	TonB-dependent receptor, plug	49 / 851	13.0	2×10^{-33}
PF00593	TonB-dependent receptor, barrel	44 / 665	14.9	2×10^{-32}
PF14322	SusD-like, glycan binding	36 / 505	16.0	9×10^{-28}
PF07980	SusD family	36 / 518	15.5	2×10^{-27}
PF02915	Ruberrythrin (oxidative stress)	12 / 65	40.4	8×10^{-14}

Table S8 — Bootstrap 95 % confidence intervals on the 650M test set ($B = 1000$ resamples). Primary metric in bold. Relates to Section 5.9.

Method	AUPRC [95 % CI]	AUROC [95 % CI]
MetaWIBELE unsupervised	0.075 [0.070, 0.082]	0.730 [0.720, 0.739]
MetaWIBELE supervised	0.064 [0.059, 0.070]	0.703 [0.694, 0.712]
Abundance MLP	0.142 [0.133, 0.152]	0.831 [0.825, 0.838]
Sequence ESM-2 650M	0.271 [0.257, 0.286]	0.910 [0.905, 0.914]
Fusion (abundance + ESM-2)	0.368 [0.353, 0.385]	0.927 [0.923, 0.931]

Method	AUPRC [95 % CI]	AUROC [95 % CI]
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Table S9 — The leakage trap (ElasticNet, HMP2 species, IBD vs control). Relates to Section 6.2.

Evaluation	AUROC
HMP2 naive sample-level 5-fold CV (leaky)	0.948
HMP2 participant-grouped 5-fold CV (correct)	0.460

Table S10 — EC-enzyme cross-cohort transfer, AUROC (2,052 shared EC numbers). Relates to Section 6.5.

Model	within-cohort CV	HMP2 → Franzosa	Franzosa → HMP2
ElasticNet	0.905	0.704	0.690
PCA32 + LR	0.901	0.704	0.667
Autoencoder32 + LR	0.908	0.738	0.653
VAE32 + LR	0.867	0.687	0.629

Table S12 — Sample-level dysbiosis classification, all six method–feature pairings (patient-level held-out test set). Relates to Section 3.2.

Method	Features	test AUROC	test AUPRC	test F1
ElasticNet	species	0.857	0.694	0.662
ElasticNet	pathways	0.940	0.866	0.733
PCA32 + LR	species	0.612	0.507	0.490
PCA32 + LR	pathways	0.885	0.738	0.654
AE32 + LR	species	0.495	0.407	0.349
AE32 + LR	pathways	0.809	0.655	0.581

Table S13 — Clean accuracy and robustness to corrupted inputs (test AUROC; dysbiosis, community pathways). Relates to Section 3.4.

Model	clean	noise $\sigma = 3$	70 % missing
ElasticNet	0.963	0.765	0.835
PCA32 + LR	0.951	0.871	0.919
Autoencoder32 + LR	0.964	0.866	0.911
Denoising AE32 + LR	0.951	0.892	0.921
VAE32 + LR	0.965	0.892	0.933

Table S11 — PCA of the pooled cohorts (197 shared species) and how separable each structure is along the first ten components. Relates to Section 6.7.

Quantity	Value
PC1 variance explained	8.9 %
PC2 variance explained	5.8 %
10-PC logistic AUROC, cohort (batch)	0.99
10-PC logistic AUROC, diagnosis (biology)	0.66

Supplementary Figures

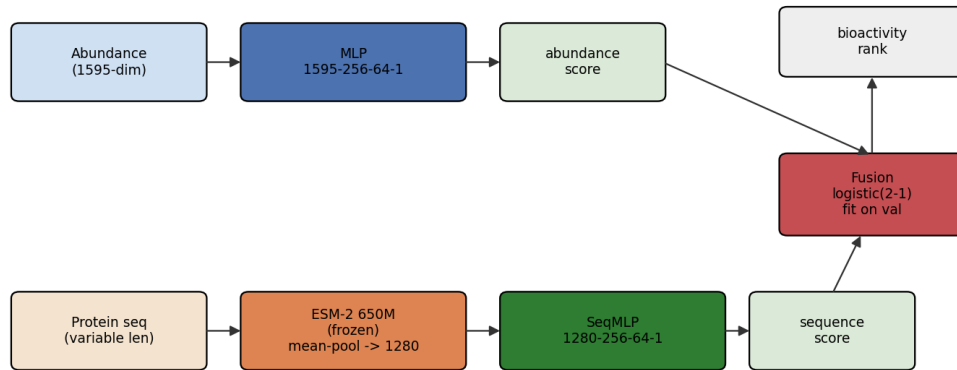


Figure S1 — Fusion architecture: the abundance MLP and the frozen ESM-2 sequence encoder each produce a score, combined by a validation-fitted logistic calibrator. Relates to Section 2.3.

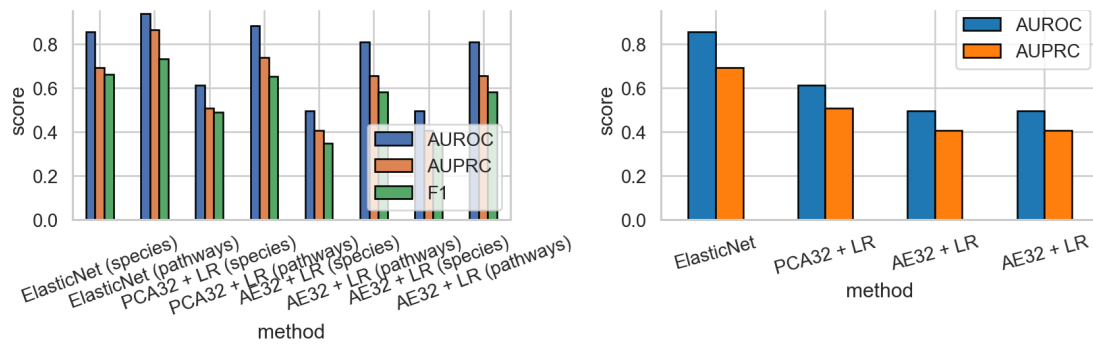


Figure S2 — Sample-level phenotype classification, test-set performance. Left: AUROC, AUPRC and F1 for every method across both feature sets (species abundances and community pathways). Right: AUROC and AUPRC restricted to the species feature set. Relates to Section 3.2.

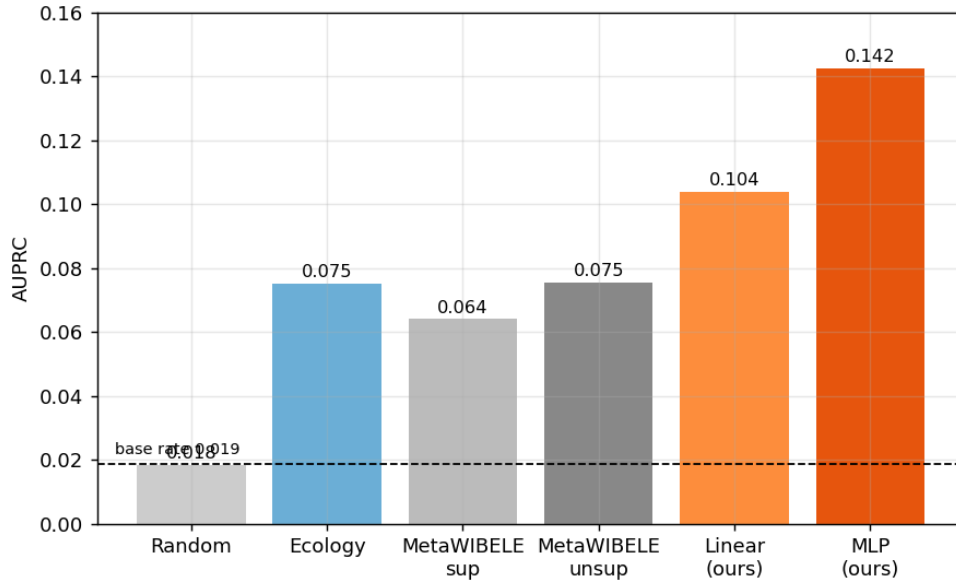


Figure S3 — Protein-family prioritization: AUPRC by method on the test set. The MLP nearly doubles the published MetaWIBELE baseline. Relates to Section 4.2.

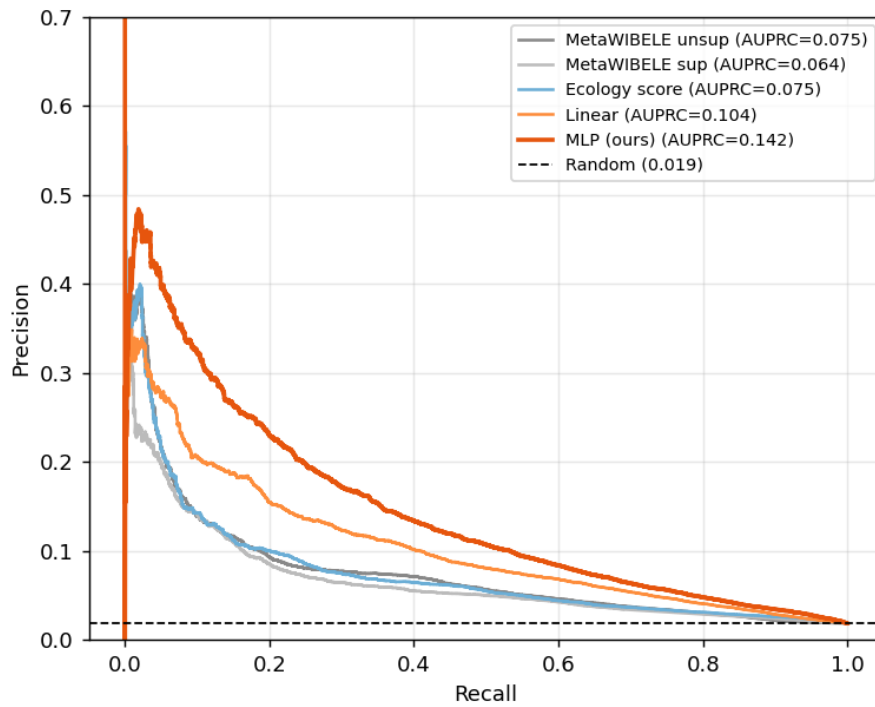


Figure S4 — Protein-family prioritization: precision-recall curves. The MLP (ours) stays above the baselines across the operating range. Relates to Section 4.2.

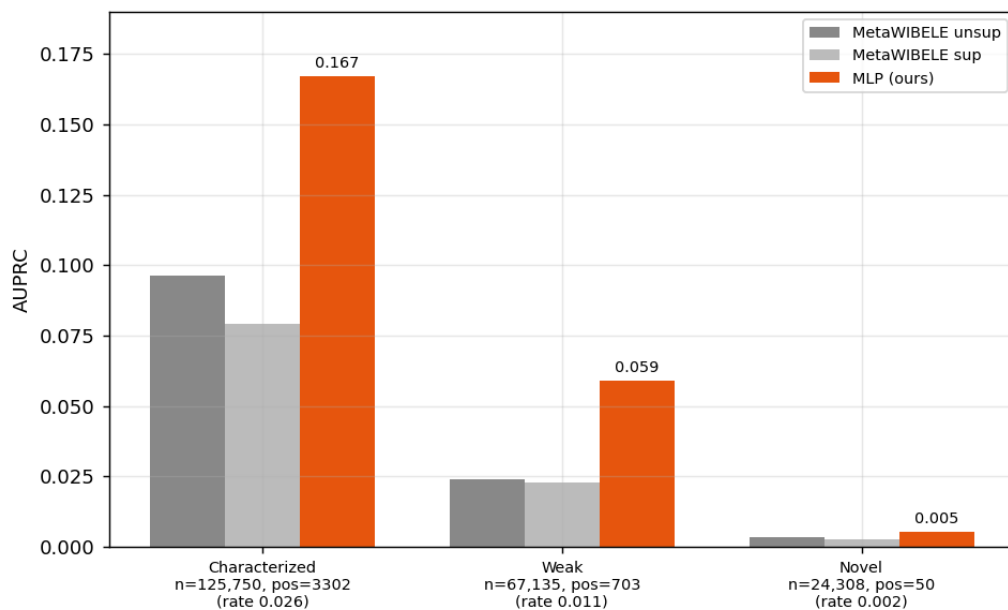


Figure S5 — Protein-family prioritization: AUPRC by characterization category. The MLP wins in every stratum, with the largest relative gain on weak-homology families. Relates to Section 4.3.

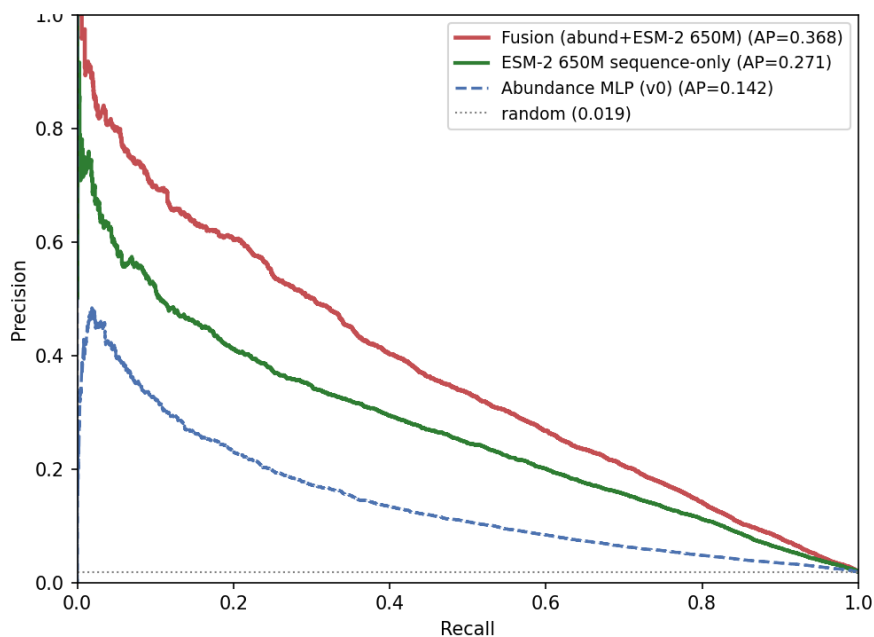


Figure S6 — Sequence and fusion models: precision-recall curves for the ESM-2 650M sequence-only and fusion models versus the abundance baseline (test). Relates to Section 5.2.

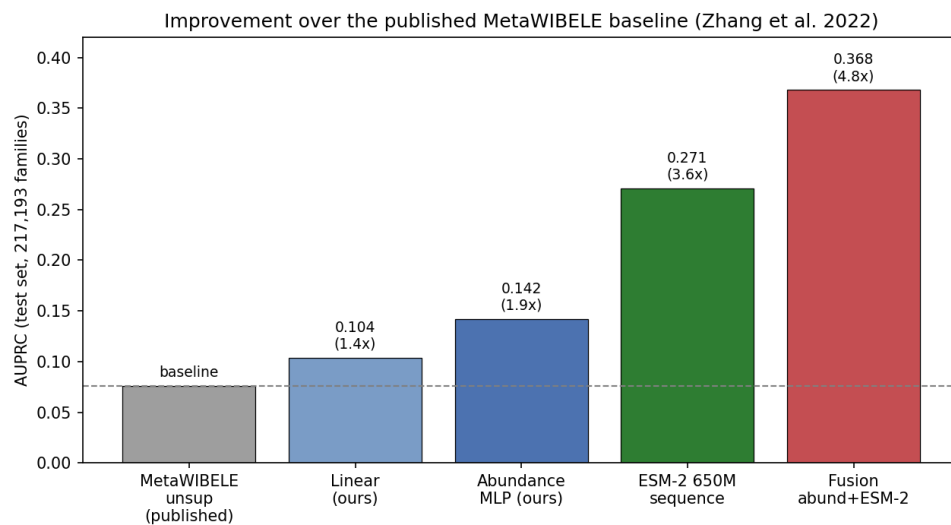


Figure S7 — Improvement over the published MetaWIBELE baseline: AUPRC of each model with the relative factor annotated. The dashed line is the MetaWIBELE baseline (0.076). Relates to Section 5.4.

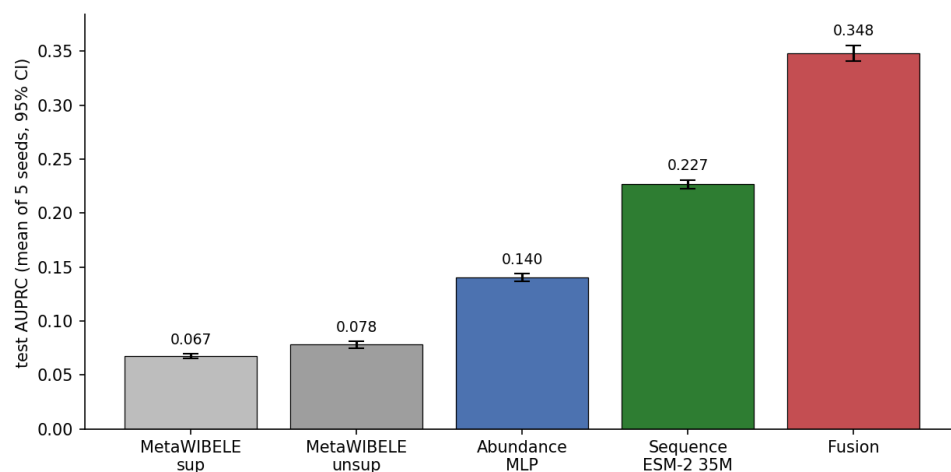


Figure S8 — Prioritization robustness: mean test AUPRC over five random splits with 95 % confidence intervals. Every model's interval sits well clear of MetaWIBELE's. Relates to Section 5.5.

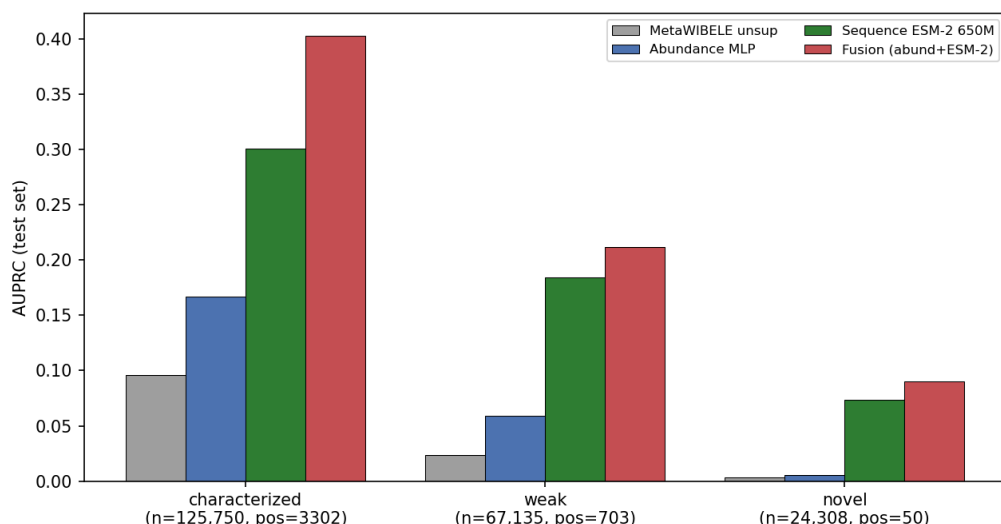


Figure S9 — Sequence model by characterization bucket: the ESM-2 sequence model lifts the weak and novel strata far above the abundance model, which is where annotation-based methods struggle most. Relates to Section 5.6.

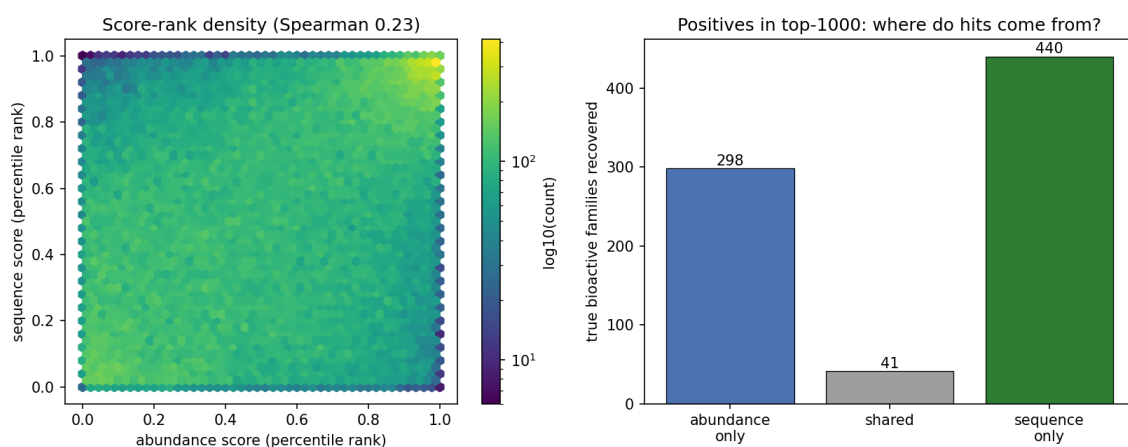


Figure S10 — Left: density of the two score ranks (weak correlation). Right: of the true positives recovered in the top 1,000, most are unique to either the abundance or the sequence model rather than shared. Relates to Section 5.7.

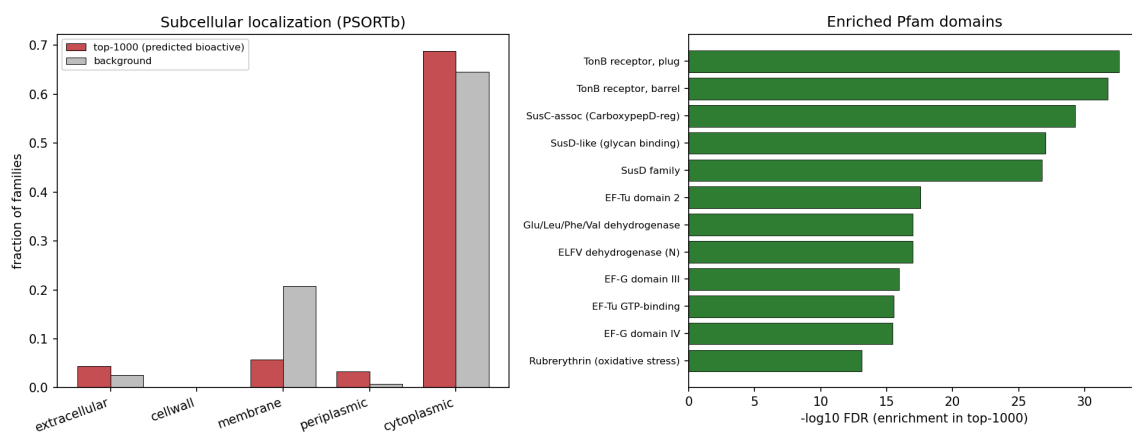


Figure S11 — Interpretation of the prioritized families. Left: the top-ranked families are enriched for extracellular and periplasmic proteins, not inner-membrane ones. Right: the most enriched Pfam domains are the SusC/SusD glycan-foraging system, with elongation factors and central metabolism also present. Relates to Section 5.8.

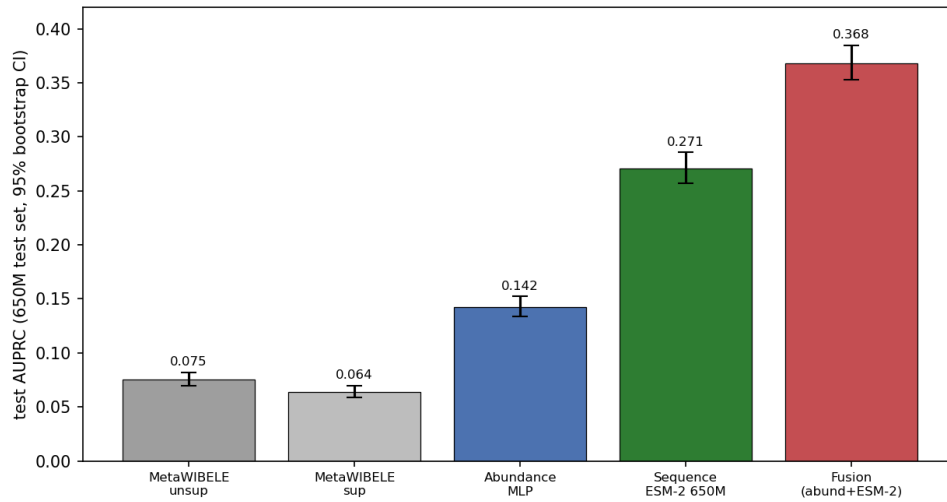


Figure S12 — Headline prioritization metrics with 95 % bootstrap confidence intervals on the fixed 650M test set. The intervals are narrow and do not overlap between consecutive methods, so each rung of the ladder is a statistically clear improvement. Relates to Section 5.9.

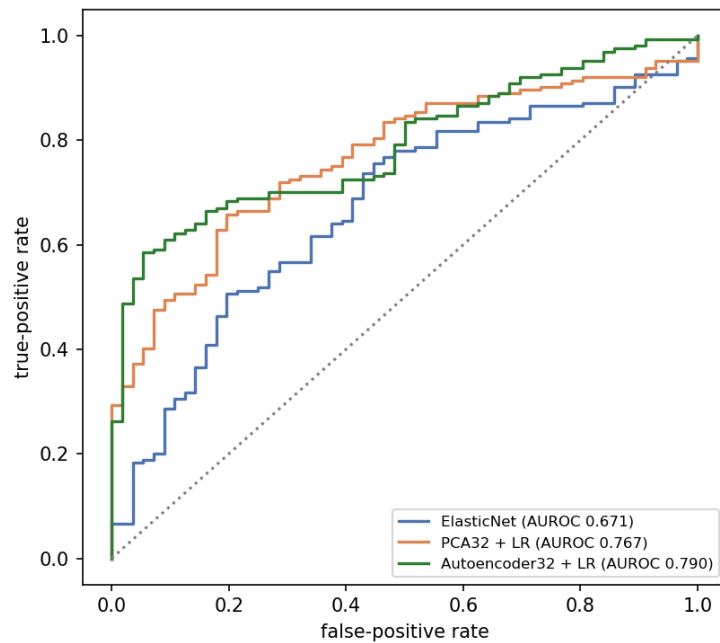


Figure S13 — Cross-cohort ROC curves for the HMP2-trained models applied to the independent Franzosa cohort. Relates to Section 6.3.

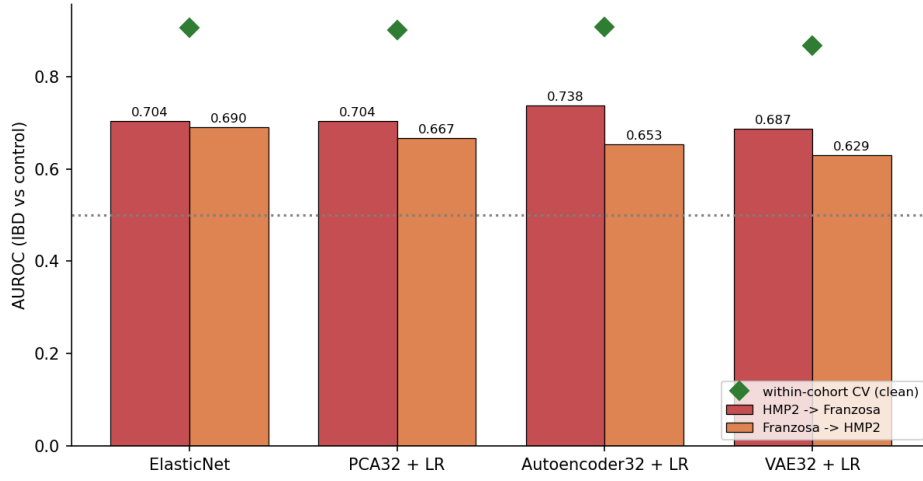


Figure S14 — High-dimensional companion: cross-cohort transfer on 2,052 shared EC enzymes. The autoencoder again transfers best in the data-rich direction. Relates to Section 6.5.

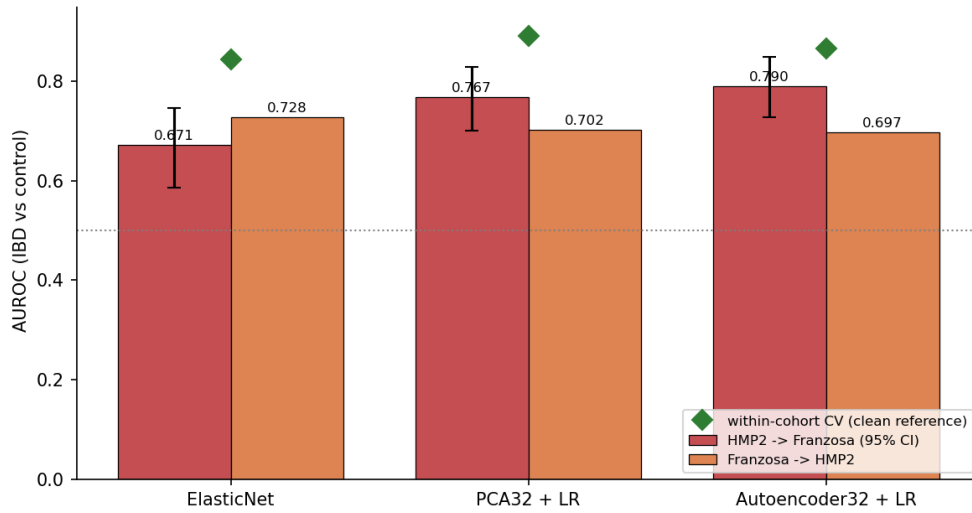


Figure S17 — Cross-cohort transfer AUROC in both directions, with the clean within-cohort CV as a reference diamond and 95 % bootstrap CIs on the HMP2 $\rightarrow$ Franzosa bars. Training on the larger HMP2 and testing on Franzosa, the learned representations transfer best. Relates to Section 6.3.

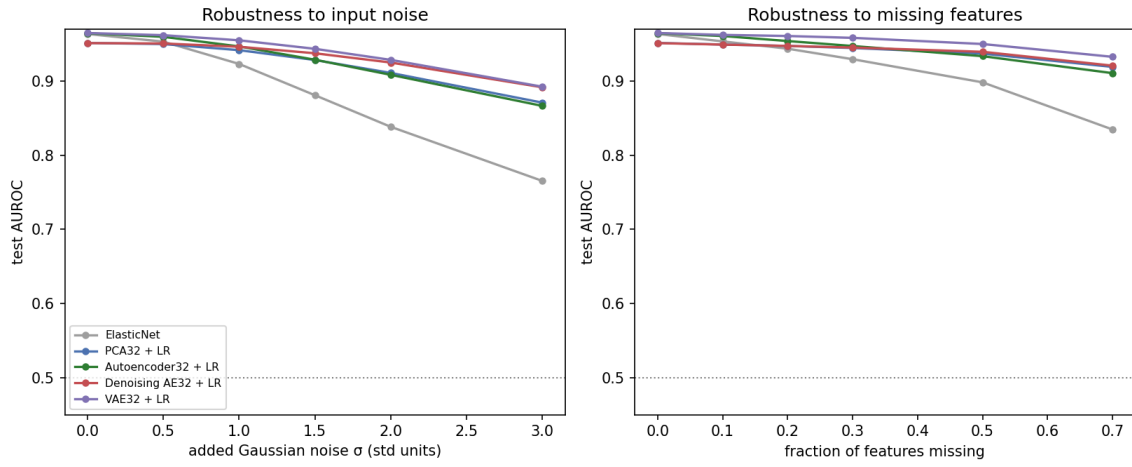


Figure S16 — Robustness of the sample-level classifier: test AUROC as the inputs are corrupted with Gaussian noise (left) and random missingness (right). The linear ElasticNet degrades fastest; the learned low-dimensional codes, and the VAE most of all, are far more robust. Relates to Section 3.4.

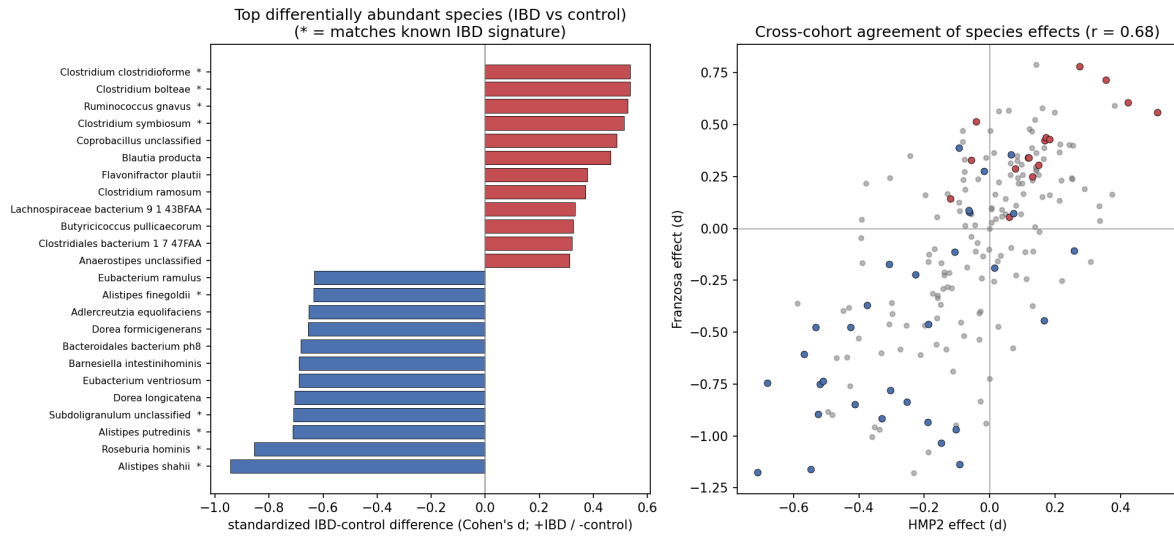


Figure S15 — Cross-cohort interpretation. Left: top differentially abundant species (IBD vs control), * marks those matching the known IBD signature. Right: the two cohorts agree on species effects ($r = 0.68$), coloured points are known IBD-up (red) and IBD-down (blue) taxa. Relates to Section 6.6.